

But what if an investor cares about other things? In particular, what if the investor cares about downside risk and other higher moment measures of risk?

Variances always decrease when assets with nonperfect correlations are combined. This causes improvements in returns and reductions in risk in mean-variance space. But other measures of risk do not necessarily diminish when portfolios are formed. For example, a portfolio can be more negatively skewed and thus have greater downside risk than the downside risk of each individual asset.¹⁸ Most investors care about many more risk measures than simply variance.

Diversification is not necessarily a free lunch when other measures of risk are considered. Nevertheless, from the viewpoint of characterizing the tails of asset returns, standard deviation (or variance) is the most important measure. Furthermore, optimal asset weights for a general utility function can be considered to be mean-variance weights to a first approximation, but we may have to change an investor's risk aversion in the approximation (see chapter 2).¹⁹ While variance is the first-order risk effect, in some cases the deviations from the mean-variance approximation can be large. You still need to watch the downside.

In the opposite direction, diversification kills your chances of the big lottery payoff. If you are risk seeking and want to bet on a stock having a lucky break—and hope to become a billionaire from investing everything in the next Microsoft or Google—diversification is not for you. Since diversification reduces idiosyncratic risk, it also limits the extremely high payoffs that can occur from highly concentrated positions. The risk-averse investor likes this because it also limits the catastrophic losses that can result from failing to diversify. Just ask the employees of Enron and Lehman and the faculty and students at the University of Rochester and Boston University.

The overall message from mean-variance investing is that diversification is good. It minimizes risks that are avoidable and idiosyncratic. It views assets holistically, emphasizing how they interact with each other. By diversifying, investors improve their Sharpe ratios and can hold strictly better portfolios—portfolios that have higher returns per unit of risk or lower risk for a given target return—than assets held in isolation.

3. Mean-Variance Optimization

We've described the mean-variance frontier and know that the best investment opportunities lie along it. Which of these efficient portfolios on the mean-variance frontier should we pick?

¹⁸ The technical jargon for this is that higher moment risk measures are not necessarily subadditive. See Artzner et al. (1999).

¹⁹ Technically, you can Taylor expand a utility function so that the first term represents CRRA utility, which is approximately mean-variance utility.